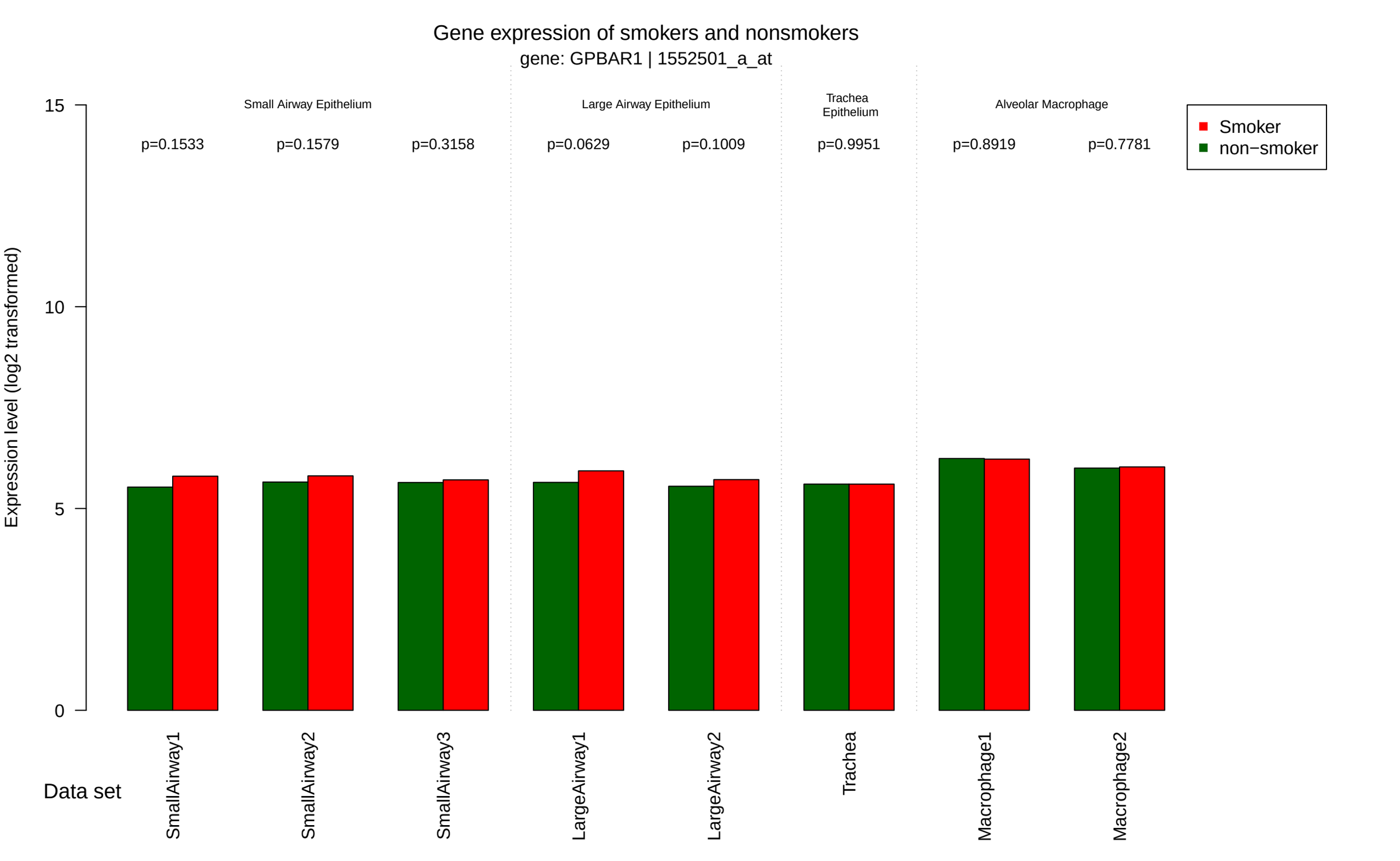
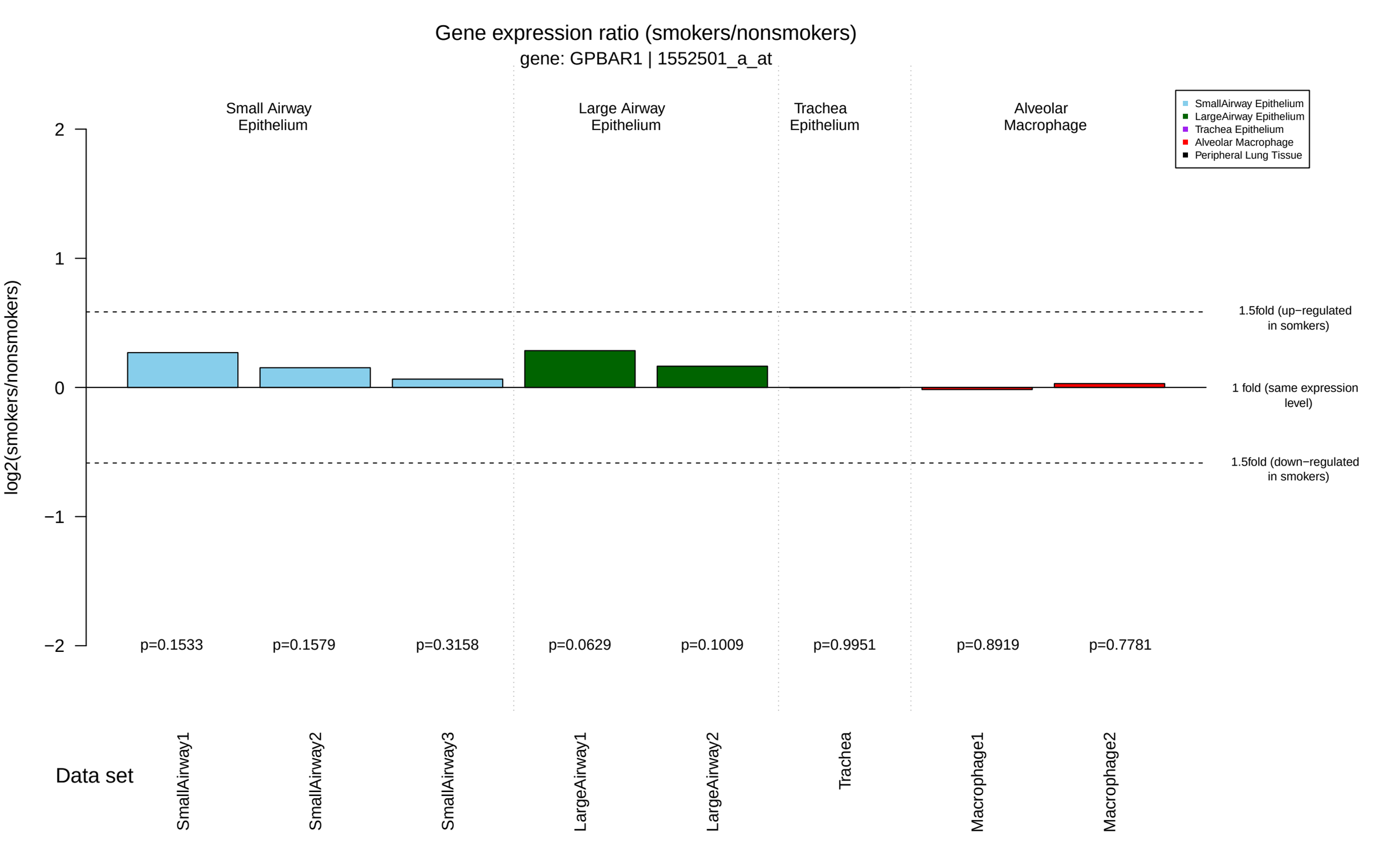
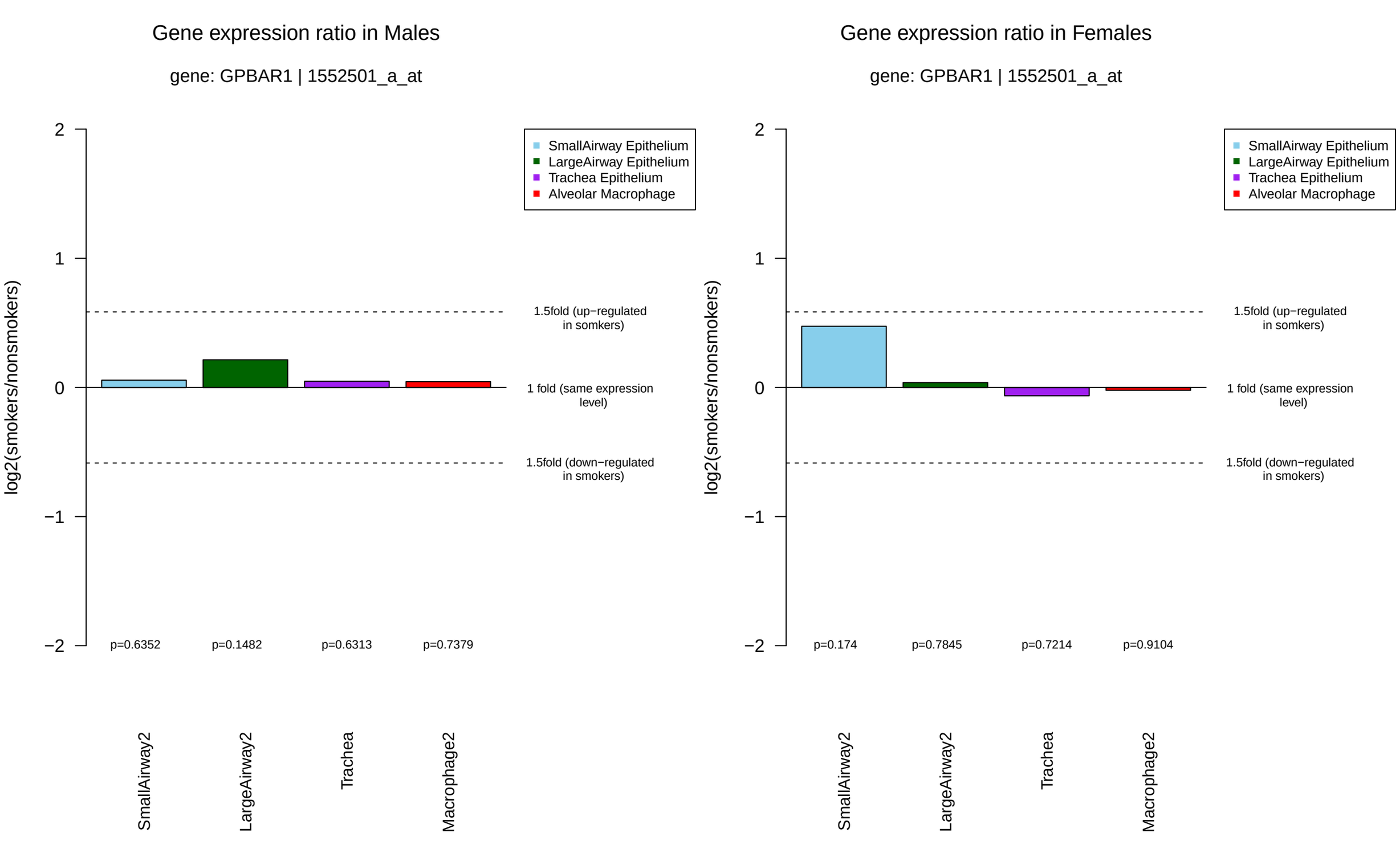
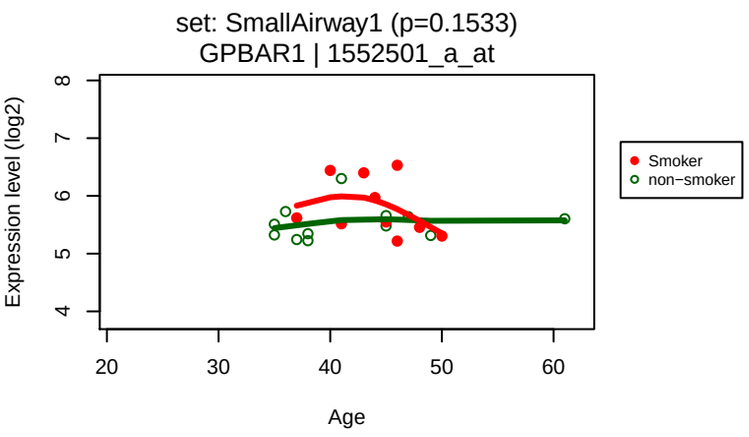
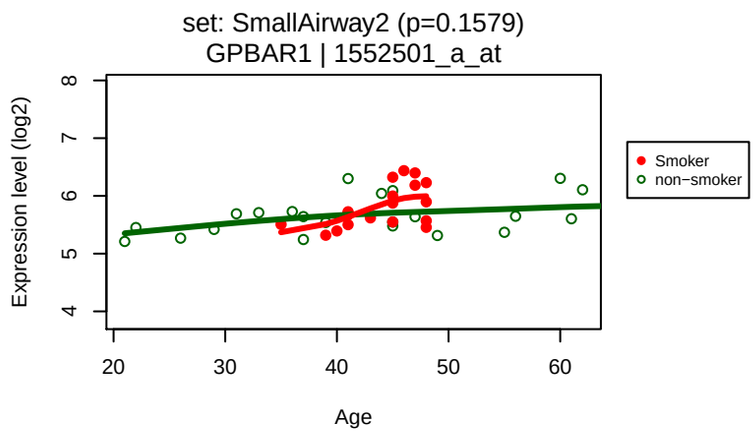
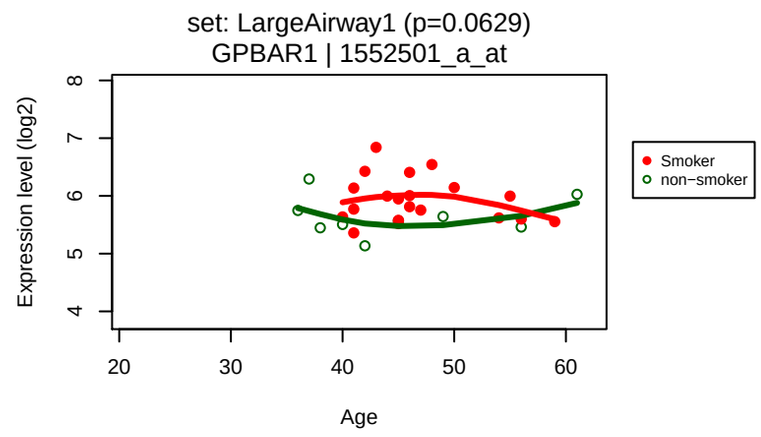

Gene expression of smokers and nonsmokers
gene: GPBAR1 | 1552501_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1533
p=0.1579
p=0.3158
p=0.0629
p=0.1009
p=0.9951
p=0.8919
p=0.7781
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

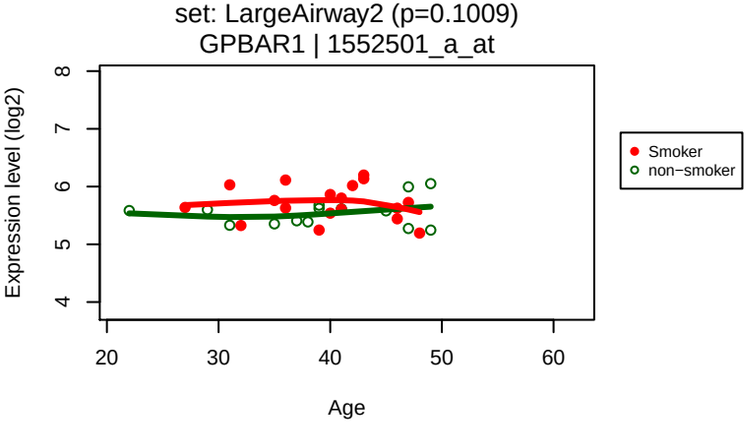

Gene expression ratio (smokers/nonsmokers)
gene: GPBAR1 | 1552501_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1533
p=0.1579
p=0.3158
p=0.0629
p=0.1009
p=0.9951
p=0.8919
p=0.7781
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

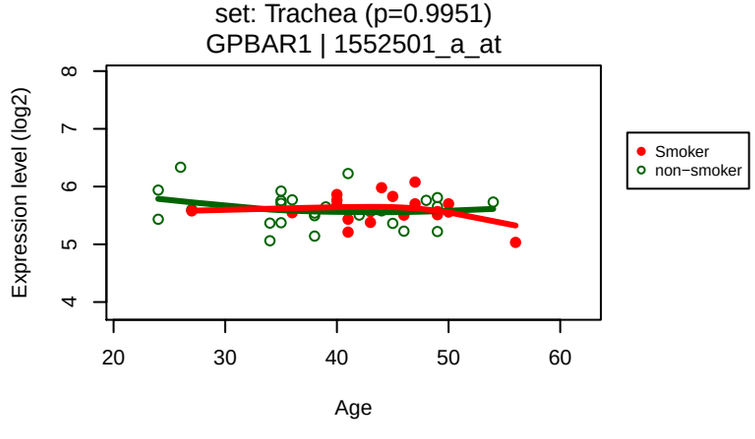

Gene expression ratio in Males
gene: GPBAR1 | 1552501_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6352
p=0.1482
p=0.6313
p=0.7379
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPBAR1 | 1552501_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.174
p=0.7845
p=0.7214
p=0.9104
SmallAirway2
LargeAirway2
Trachea
Macrophage2

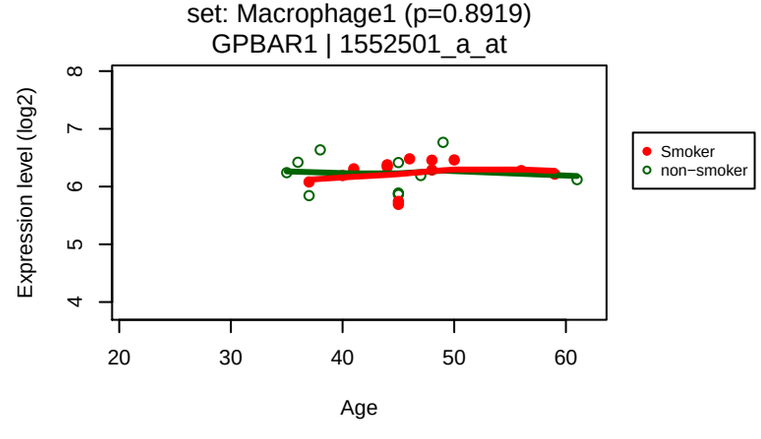

set: SmallAirway1 (p=0.1533)
GPBAR1 | 1552501_a_at
Expression level (log2)
Age
Smoker
non−smoker

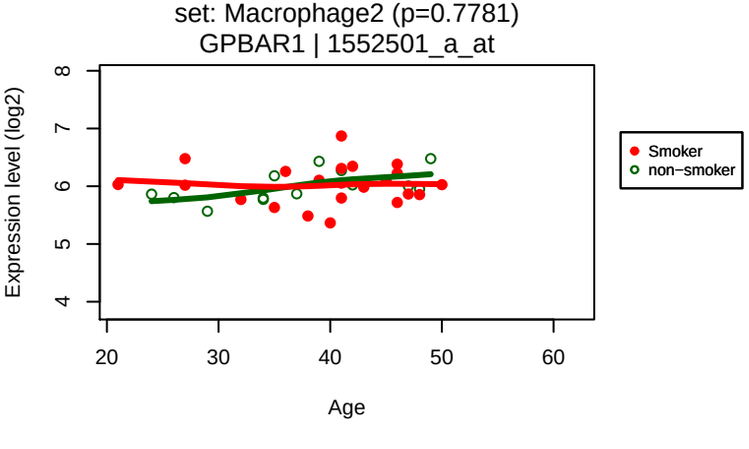

set: SmallAirway2 (p=0.1579)
GPBAR1 | 1552501_a_at
Expression level (log2)
Age
Smoker
non−smoker

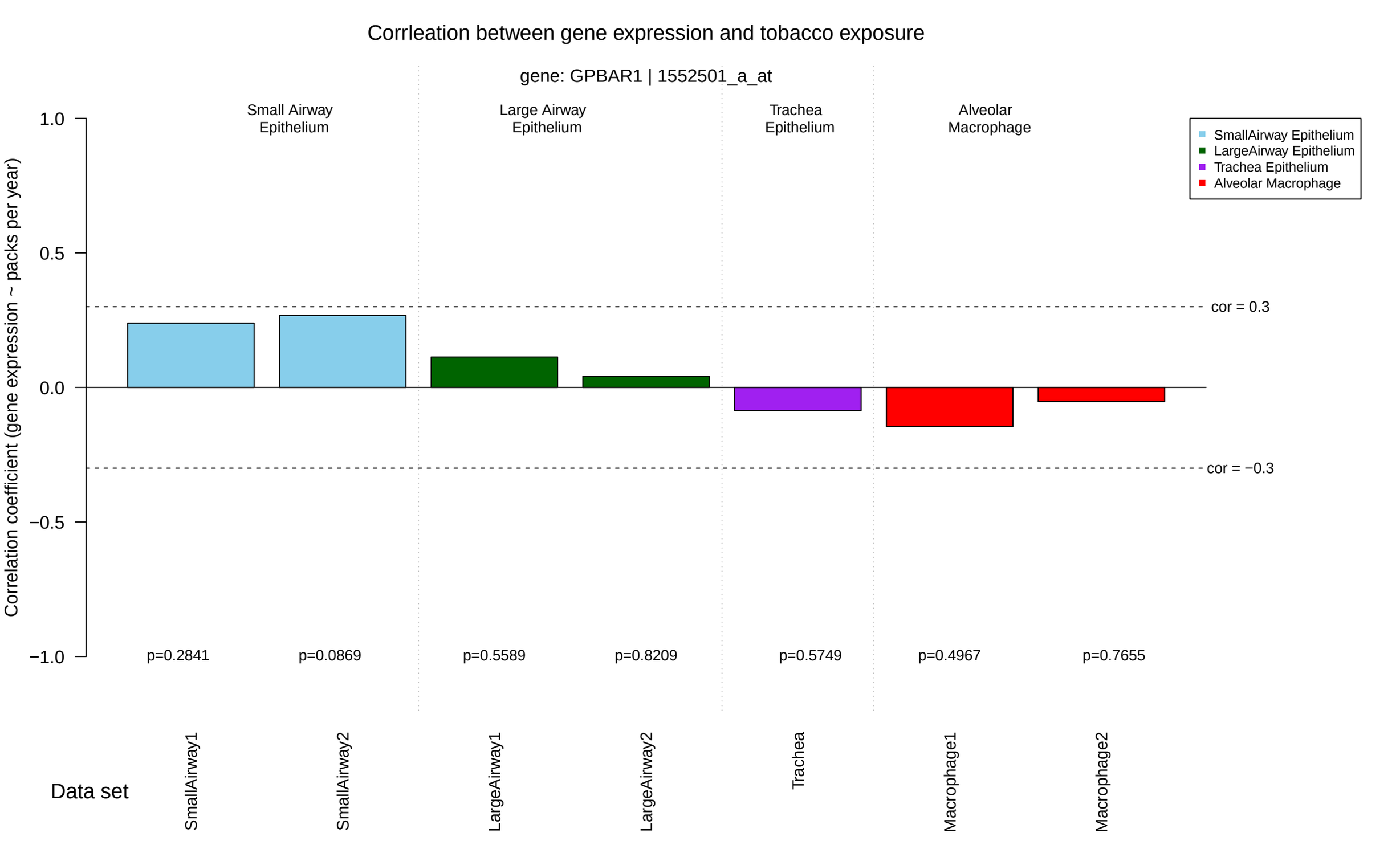

set: LargeAirway1 (p=0.0629)
GPBAR1 | 1552501_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1009)
GPBAR1 | 1552501_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9951)
GPBAR1 | 1552501_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8919)
GPBAR1 | 1552501_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7781)
GPBAR1 | 1552501_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPBAR1 | 1552501_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
Correlation coefficient (gene expression ~ packs per year)
cor = 0.3
cor = −0.3
p=0.2841
p=0.0869
p=0.5589
p=0.8209
p=0.5749
p=0.4967
p=0.7655
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2